

CS23334 Fundamentals of Data Science Lab

Experiment 2

Exp No:2 Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

Description: Use sample data set sales-data.csv.

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the data into a pandas DataFrame
file_path='C:\sales_data.csv'
df = pd.read_csv(file_path)

# Display the first few rows of the DataFrame
print(df.head())

# Check for missing values
print(df.isnull().sum())

# Fill or drop missing values if necessary
df['Sales'].fillna(df['Sales'].mean(), inplace=True)
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)

# Summary statistics
print(df.describe())

# Group by product and calculate the total sales and quantity
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()
print(product_summary)
```

Code Continued:

```
# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()

# Line plot of sales over time
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date').agg({'Sales': 'sum'}).reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.show()

# Pivot table to analyze sales by region and product
pivot_table = df.pivot_table(values='Sales', index='Region', columns='Product',
                              aggfunc=np.sum, fill_value=0)
print(pivot_table)

# Correlation matrix
correlation_matrix = df.corr()
print(correlation_matrix)

# Heatmap of the correlation matrix
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

OUTPUT:

```
Date      Product  Sales  Quantity  Region
0 2023-01-01 Product A    200         4   North
1 2023-01-02 Product B    150         3   South
2 2023-01-03 Product A    220         5   North
3 2023-01-04 Product C    300         6   East
4 2023-01-05 Product B    180         4   West
```

```
Date      0
Product    0
Sales      0
Quantity   0
Region     0
dtype: int64
```

```
           Sales  Quantity
count  16.000000  16.000000
mean   237.500000  5.375000
std     64.031242  1.746425
min    150.000000  3.000000
25%    187.500000  4.000000
50%    225.000000  5.500000
75%    302.500000  7.000000
max     340.000000  8.000000
```

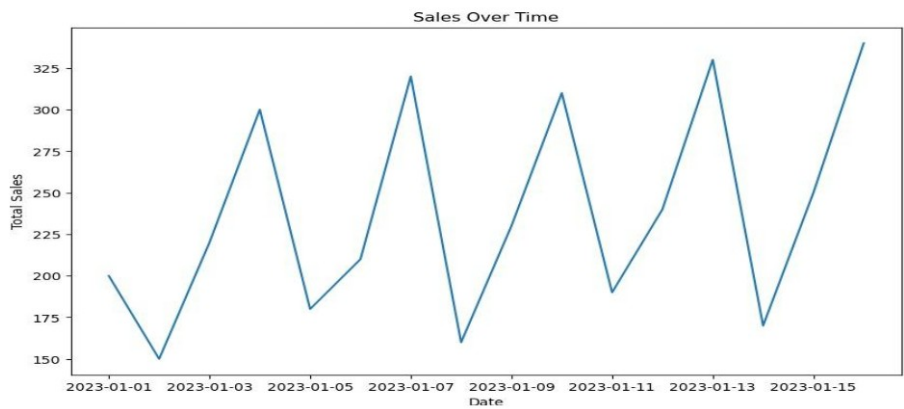
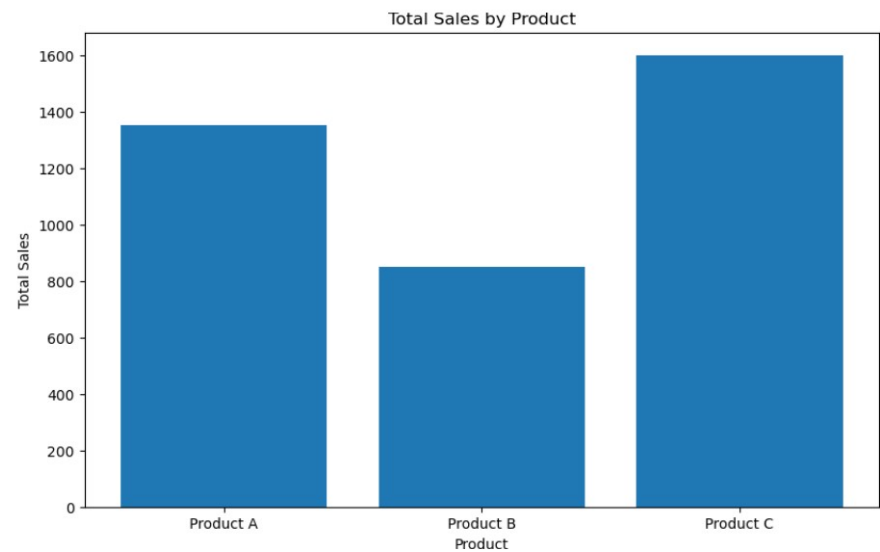
```
           Product  Sales  Quantity
0  Product A     1350        33
1  Product B      850        17
2  Product C     1600        36
```

```
Product  Product A  Product B  Product C
Region
East           0           0         1600
North         1350           0           0
South           0          480           0
West           0          370           0
```

```
           Sales  Quantity
Sales      1.000000  0.944922
Quantity   0.944922  1.000000
```

```
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_9648\511106317.py:49: FutureWarning: The default
value of numeric_only in DataFrame.corr is deprecated. In a future version, it will default to
False. Select only valid columns or specify the value of numeric_only to silence this warning.
correlation_matrix = df.corr()
```

Output Figures



Output Figure Continued

